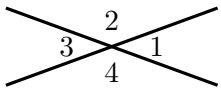
Proving Angle Relationships — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Vertical Angles and Linear Pairs

Watch Me — Worked Example



$m\angle 1 = 48°$. $\angle 3$ is vertical to $\angle 1$: $m\angle 3 = 48°$.
$\angle 2$ and $\angle 1$ are a linear pair: $m\angle 2 = 180° - 48° = 132°$; $m\angle 4 = 132°$ (vertical to $\angle 2$).

1. Use the figure above. $m\angle 2 = 115°$. Find $m\angle 4$.

$115°$

2. $m\angle 2 = 115°$. Find $m\angle 1$.

$65°$

3. $m\angle 3 = 72°$. Find $m\angle 1$, $m\angle 2$ and $m\angle 4$.

$72°,\ 108°,\ 108°$

4. Are $\angle 1$ and $\angle 2$ vertical angles or a linear pair?

a linear pair

Type 2 Solve With Algebra

Watch Me — Worked Example

$m\angle 1 = 4x + 6$ and $m\angle 3 = 6x - 14$ (vertical angles). $4x + 6 = 6x - 14 \Rightarrow 20 = 2x \Rightarrow x = 10$.
$m\angle 1 = 4(10) + 6 = 46°$. **Answer: $x = 10$, $m\angle 1 = 46°$**

5. $m\angle 1 = 5x - 12$, $m\angle 3 = 3x + 20$. Find x and $m\angle 1$.

$x = 16, m\angle 1 = 68°$

6. $m\angle 1 = 2x + 10$, $m\angle 2 = 4x - 10$ (linear pair). Find x and $m\angle 2$.

$x = 30, m\angle 2 = 110°$

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7. $\angle A$ and $\angle B$ are both complementary to $\angle C$. $m\angle A = 3x - 4$, $m\angle B = x + 20$. Find $m\angle A$.

$x = 12$, $m\angle A = 32°$

8. HARD $m\angle 1 = x^2$, $m\angle 3 = 5x + 14$ (vertical angles), $x > 0$. Find $m\angle 1$.

$x = 7$, $m\angle 1 = 49°$

Type 3 Complete the Proof

Watch Me — Worked Example

Given: $\angle 1$ and $\angle 2$ are right angles. Prove: $\angle 1 \cong \angle 2$.
1. $\angle 1$ and $\angle 2$ are right angles (Given) 2. $\angle 1 \cong \angle 2$ (Right Angles Congruence Theorem)

9. Given: two lines intersect, forming $\angle 1$–$\angle 4$ as in the Type 1 figure.
Prove: $\angle 2 \cong \angle 4$.

Statements	Reasons
1. $\angle 2$ and $\angle 4$ are vertical angles	Definition of vertical angles
2. $\angle 2 \cong \angle 4$	Vertical Angles Congruence Theorem

10. Given: $\angle 1$ and $\angle 2$ are a linear pair, and $\angle 2$ and $\angle 3$ are a linear pair. Prove: $\angle 1 \cong \angle 3$.

Statements	Reasons
1. $\angle 1, \angle 2$ and $\angle 2, \angle 3$ are linear pairs	Given
2. $\angle 1, \angle 2$ supplementary; $\angle 2, \angle 3$ supplementary	Linear Pair Postulate
3. $\angle 1 \cong \angle 3$	Congruent Supplements Theorem

Type 4 Always, Sometimes or Never?

Watch Me — Worked Example

"Two angles that form a linear pair are congruent." Only when both are $90°$. **Answer: sometimes.**

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11. Vertical angles are congruent.

always

12. Two right angles are supplementary.

always

13. Two supplementary angles are both acute.

never

14. HARD Two angles complementary to the same angle are congruent.

always

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